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Optimizing straw anaerobic digestion via nanobubble technology: Insights into hydrolysis, acidogenesis, and methanogenesis processes

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ABSTRACT

Anaerobic digestion (AD) is an effective approach for the resource and energy recovery of straw biomass. However, the process faces challenges such as poor hydrolysis efficiency, slow startup, and lower overall efficiency. Nanobubble technology offers a promising solution to enhance methane production without causing secondary pollution. This study investigates the effects of nanobubble water (NBW) prepared using different gases (N_2-NBW, O_2-NBW, H_2-NBW, CO_2-NBW, and Air-NBW) on the methane production characteristics and microbial communities during AD of corn straw. The results showed that methane production increased by 11.86 %–24.08 % compared to the control group, with the maximum cumulative methane yield reaching 189.36 ± 5.32 mL/g VS. The modified Gompertz equation effectively fitted the cumulative methane production curves, with a high correlation coefficient ($R^2 > 0.99$). Total solids (TS) and volatile solids (VS) removal rates improved by 9.70 %–26.28 % and 11.87 %–24.09 %, respectively. Microbial community analysis revealed an increased relative abundance of bacteria such as *Ruminofilibacter*, which facilitates hydrolysis and acidogenesis. Additionally, the relative abundance of methanogenic archaea such as *Methanosarcina* and *Methanobacterium* also increased to varying degrees. Consequently, the addition of NBW promotes the processes of hydrolysis, acetogenesis, and methanogenesis, providing new opportunities for the efficient utilization of straw biomass.

1. Introduction

With growing global energy demand and increasing severity of environmental issues, the development of renewable energy has become a research hotspot in today's society [1]. Straw, as a widely distributed agricultural waste, has an annual production of approximately 1 billion tons in China [2]. Its resource utilization through AD can not only reduce environmental pollution but also convert it into cleaner biogas, alleviating the dependence on fossil fuels and reducing carbon emissions [3,4]. However, the efficient bioconversion of straw is hindered by its pronounced recalcitrance, which stems from the complex physicochemical structure of lignocellulosic biomass. At the structural level, cellulose, hemicellulose, and lignin are tightly interwoven to form a rigid plant cell wall, creating a natural barrier to microbial and enzymatic attack [5]. The high crystallinity of cellulose and the presence of extensive hydrogen bonding networks reduce the accessibility of hydrolytic enzymes, while hemicellulose acts as a cross-linking matrix that further restricts degradation. Lignin, with its highly hydrophobic and aromatic polymeric structure, not only provides physical rigidity but also exerts non-productive binding with enzymes, thereby inhibiting hydrolysis. In addition, the low porosity and limited surface area of straw cell walls hinder the penetration of microbial consortia and enzymatic systems into the substrate. Collectively, these physicochemical barriers make hydrolysis the rate-limiting step of straw anaerobic digestion, emphasizing the need for effective pretreatment and enhancement strategies to overcome recalcitrance and improve overall bioconversion efficiency [6]. In addition, maintaining a dynamic balance between hydrolysis and methanogenesis during anaerobic

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digestion is a key factor for improving substrate conversion efficiency and methane yield [7].

In recent years, many technologies have been developed to increase biogas production from straw through AD, aiming to overcome its complex and recalcitrant lignocellulosic structure [8]. Nowadays, the most common processes for improving lignocellulosic feed-stock degradation through AD are chemical (alkaline and acidic), biological (fungal or enzyme), oxidative(wet oxidation and wet explosion), and physical (milling, extrusion, ultrasonic, steam explosion, and hydrothermal treatment) [9]. Even if the lignocellulose's degradability is improved following the aforementioned treatments, each pretreatment technique has some pros and cons. For instance, physical pretreatments such as milling or steam explosion typically require energy inputs ranging from 0.5 to 5 MJ/kg of dry biomass, while chemical pretreatments may achieve high cellulose conversion rates (up to 70–80 %) but often produce inhibitory compounds such as furfural or generate wastewater requiring further treatment [10]. Biological pretreatments are generally energy-efficient and environmentally friendly but need prolonged processing times (weeks to months) to achieve moderate lignin removal (10–30 %). These limitations call for the exploration of more effective, energy-efficient, and less chemical-intensive pretreatment methods. Recent advances in biomass utilization technologies provide valuable context for such innovations. For example, Sun et al. reported a concerted catalysis strategy for bio-ethanol activation and dehydrogenation, demonstrating how coordinated chemical and catalytic approaches can significantly enhance biomass conversion efficiency [11]. Building on this perspective of technological innovation, nanobubble water (NBW) pretreatment emerges as a promising, non-chemical alternative, potentially reducing energy input and chemical usage while maintaining or improving biogas yields. Therefore, NBW offers a more sustainable and efficient approach compared with existing pretreatment strategies.

NBW is a special liquid form containing a large number of nanometer-sized bubbles (with diameters less than 1 μm), generated in water through methods such as mechanical agitation [12], hydrodynamic cavitation [13], ultrasonic cavitation [14], and membrane filtration [15]. Because of this, it possesses more reactive oxygen species (ROS), a greater specific surface area, a faster mass transfer efficiency, a longer residence time, a higher surface charge, and stronger gas solubility, among other characteristics [16,17]. However, NBWs also exhibit potential instability over time, and their large-scale generation may involve operational complexity and significant energy requirements, which need to be critically considered for practical applications [18]. In recent years, NBW has been widely applied in fields such as wastewater treatment, soil improvement, aquaculture, and waste treatment [18]. Recent studies have shown that NBW also has a significant positive effect on the AD process. It can increase methane production and promote substrate degradation, while also not causing secondary pollution after use, offering excellent environmental benefits [19]. The mechanisms by which NBW formed from different gases (such as nitrogen, oxygen, hydrogen, carbon dioxide, and air) affect microorganisms during AD in a different way. For example, N_2-NBW provides an inert gas environment, reducing the inhibitory effects of dissolved oxygen on anaerobic bacteria [20]. O_2-NBW can improve the hydrolysis and acidogenesis processes in the early stages. The collapse of the bubbles may generate small amounts of reactive oxygen species (such as hydroxyl radicals), which help promote the opening or oxidation of refractory substances [21]. H_2-NBW can provide a hydrogen source, promoting the activity of hydrogen-producing bacteria and methane-producing bacteria [22]; CO_2- may participate in the carbon cycle, enhancing the activity of hydrogenotrophic methanogens and increasing the methane production rate [23]; Air-NBW's nitrogen and oxygen increase the gas exchange efficiency in the liquid phase, enhancing mass transfer efficiency [24]. However, these reported effects may be influenced by potential instability of NBWs, operational challenges, and the energy requirements for large-scale generation, which should be critically considered. Thus, it is crucial from both a theoretical and practical standpoint to investigate how different kinds of NBW affect different stages of straw AD.

Despite these promising findings, current research on the application of different NBW types in AD is still in its infancy. Most existing studies focus on single-gas NBW treatments, lacking a systematic comparison of the effects of various gas-based NBWs under identical conditions. Moreover, comprehensive evaluations of how these treatments influence the microbial community structure and functional metabolism during different AD stages are still scarce. In addition, previous work has rarely extended beyond overall performance measurements, with limited exploration of how NBWs regulate microbial community dynamics and functional metabolism across the distinct stages of hydrolysis, acidogenesis, and methanogenesis. This lack of comparative and mechanistic insights constitutes a critical knowledge gap that hampers the rational design and optimization of NBW-assisted AD systems.

To fill this knowledge gap, this study systematically investigates the impact of NBWs formed from different gases (N_2, O_2, H_2, CO_2, and air) on the anaerobic digestion of straw. Specifically, the study aims to (i) evaluate their effects on hydrolysis, acidogenesis, and methanogenesis; (ii) analyze changes in microbial community composition using high-throughput sequencing; and (iii) elucidate the functional mechanisms via predictive metabolic analysis. We hypothesize that different gas-based NBWs will exert distinct effects on the key stages of anaerobic digestion, that they will modulate microbial community composition and functional metabolism in a gas-dependent manner, and that these microbial and metabolic shifts will underpin the enhanced methane production observed in NBW-assisted systems. The ultimate goal is to provide an efficient, non-toxic, and environmentally friendly strategy for enhancing biogas production from lignocellulosic biomass.

2. Materials and methods

2.1. Substrate and inoculum

Corn straw collected from the Shangzhuang Experimental Station, Haidian District, Beijing, China, and used as a feedstock in this study. The substrate was initially sun-dried after collection, coarsely ground to a particle size of 2 mm, and then washed thoroughly to remove dust and other impurities. Subsequently, it was further dried in an oven maintained at 60 °C and finally milled to a 60-mesh particle size for subsequent use. The inoculum used in this study was collected from a continuous mesophilic (37 °C) AD reactor using cow dung and straw as feedstock. The inoculum was stored in a 4 °C refrigerator after collection for subsequent use. To eliminate the influence of residual degradable materials in the inoculum, it was placed in a 37 °C incubator and fermented until no more gas production was observed before the experiment. The physicochemical properties of the corn straw and inoculum are shown in Table 1.

Table 1
Characteristics of corn straw and inoculum.

Parameter (%)	Corn straw	Inoculum
C	41.04 ± 0.05	ND
H	5.40 ± 0.01	ND
N	0.62 ± 0.01	ND
S	0.12 ± 0.01	ND
C/N	66.21 ± 1.14	ND
Cellulose[a]	31.35 ± 1.06	ND
Hemicellulose[a]	18.16 ± 0.63	ND
Lignin[a]	15.17 ± 1.03	ND
Soluble sugars[a]	10.28 ± 0.83	ND
pH	ND	7.00 ± 0.10
TS[b]	95.77 ± 0.05	5.37 ± 0.04
VS[b]	90.48 ± 0.03	3.07 ± 0.17

Note: a: based on dry weight measurements, b: based on wet weight measurements; data in the table are mean ± standard deviation (n = 3), ND means not determined.

2.2. Preparation of NBW

NBW was generated in a nanobubble generator (ZJC-NM, Shanghai Zhongjing Environmental Protection Technology Co., Ltd.), which operates on the principle of pressurized gas dissolution, as described in previous studies [25]. The operating conditions of the generator were as follows: gas flow rate of 0.12 L/min, deionized water (DW) flow rate of 3 L/min, operating pressure of 0.38–0.40 MPa, and a running time of 10 min [25].

2.3. Batch biochemical methane potential (BMP) test

The batch AD experiments were carried out in serum bottles with a total volume of 250 mL, with an effective volume of 200 mL. The VS content was set to 5 %, and the substrate-to-inoculum ratio was 2:1. Deionized water was used as the control group(CK), and N_2-NBW, O_2-NBW, H_2-NBW, CO_2-NBW, and Air-NBW were used as experimental groups, added to the anaerobic fermentation experiments. To eliminate the biogas produced by the inoculated sludge itself, a blank control containing an equal amount of sludge was included. After loading all materials, the pH was adjusted to around 7 using HCl and NaOH, and the bottles were sealed with rubber stoppers and aluminum caps. All reactors were flushed with N_2 for 3 min, and then, based on preliminary experiments, the reactors were incubated at 35 ± 1 °C for 28 days. Each test was performed in triplicates and the average results were used. Before measuring the gas production, the bottles were manually shaken for 3 min at a fixed time each day.

2.4. Analysis methods

The pH value, TS, VS, and soluble chemical oxygen demand (SCOD) were analyzed following established standard methods(APHA, 2012) [26]. The contents of cellulose, hemicellulose, and lignin were quantified according to procedures described in previous studies [27]. The biogas composition was analyzed using a GC-2014 (Shimadzu, Japan) with a 2 m × 4 mm BP1 chromatographic column, operating at a pressure of 1.3 MPa. A hydrogen flame ionization detector was used with a flow rate of 25 mL/min. The detector temperatures were set at 40 °C and 100 °C. The detection time was 1.5 min with a 300 μL air sample. For the qualitative and quantitative analysis of VFAs (lactic acid, formic acid, acetic acid, and propionic acid) in the samples, high-performance liquid chromatography (LC-20A, Shimadzu, Japan) with the external standard method was employed. The chromatographic column used was Aminex HPX-87H(300 mm × 7.8 mm), with a mobile phase of 5 mM H_2SO_4 and a flow rate of 0.6 mL/min. The column oven was set at 40 °C, with an injection volume of 10 μL. The detector was SPD-S20A, with an absorbance wavelength of 205 nm, and the detection time was 30 min.

2.5. AD kinetic models

The cumulative methane production data were fitted using a modified Gompertz model to obtain parameters such as the lag phase, maximum methane production rate, and the fitted maximum methane yield (Equation (1)). The Logistic model is commonly used to describe the dynamic behavior of methane production, reflecting the rapid growth in the early phase, a slowdown in the middle phase, and stabilization in the later phase of gas production (Equation (2)). The first-order hydrolysis kinetic model simulates the hydrolysis process during fermentation based on the balance between organic matter degradation and methane production (Equation (3)).

$$P = P_{max} \times \exp\left(- \exp\left[\frac{R\,\max \times e}{P\,\max}(\lambda\text{-}t) + 1\right]\right) \quad (1)$$

in the equation, P represents the cumulative methane production at time t (mL/g VS), Pmax is the fitted maximum methane production (mL/g VS), t is the time (days), Rmaxis the maximum methane production rate (mL/g VS/d), λ represents the lag phase (days), and e is the constant 2.718.

$$P_t = \frac{P_{max}}{1 + exp\,[-k(t - t_0)]} \quad (2)$$

in the equation, P_t represents the cumulative gas production at time t; Pmax is the maximum cumulative gas production; k is the growth rate constant; and t_0 is the time required to reach the maximum growth rate.

$$K \times t = -ln\left[1 - \frac{M(t)}{M_{max}}\right] \quad (3)$$

in the equation, K is the hydrolysis rate constant (d^{-1}), t is the fermentation time (days), M(t) is the cumulative methane production on day t (mL/g VS), and M_{max} is the potential maximum methane production (mL/g VS).

2.6. Microbial community analysis

The samples were dispatched to Shanghai Meiji Biotechnology Co., Ltd. for analysis, where high-throughput sequencing of the 16S rRNA gene was subsequently conducted on the Illumina MiSeq platform. The bacterial amplification primers were 338F and 806R, and the archaeal amplification primers were 524F10extF and Arch958RmodR. A series of microbial-related analyses were conducted using the Meiji Cloud platform, and PICRUSt2 was employed to predict microbial metabolic functions and identify key enzymes [28]. Functional annotation of the results was carried out using the KEGG database (https://www.genome.jp/kegg/). Finally, the raw data generated from the microbial sequencing was uploaded to the NCBI database (https://www.ncbi.nlm.nih.gov), with the accession number PRJNA1169782.

2.7. Statistical analysis

Data statistics were performed using Excel 2010 (Microsoft Corp., Redmond, USA), and data processing and analysis were conducted using SPSS 25.0 (IBM, USA), with statistical significance set at $p < 0.05p < 0.05$. Model plotting was done using Origin 2021 (OriginLab Corp., Northampton, USA).

3. Results and discussion

3.1. Effect of NBW on methane production

Methane production is commonly considered an important indicator for evaluating the effectiveness of AD [27]. As shown in Fig. 1a, the gas production process from corn straw exhibits three peak periods, corresponding to the effective utilization of soluble organic matter (2 days), cellulose (8 days), and hemicellulose and other substances (18 days). Fig. 1b displays the methane contents of all samples, with overall differences in methane content among the different NBW treatment groups being relatively small. Fig. 1c shows the cumulative methane production curve, clearly indicating that the addition of NBW significantly increased methane production ($p < 0.05$). Specifically, the cumulative methane production in the N_2-NBW, O_2-NBW, H_2-NBW, CO_2-NBW, and Air-NBW treatment groups was 176.94 ± 3.27, 170.72 ± 5.14, 186.17 ± 3.31, 189.36 ± 5.32, and 179.95 ± 5.22 mL/g VS, respectively, which is an increase of 15.94 %, 11.86 %, 21.99 %, 24.08 %, and 17.92 % compared to the CK (DW, 152.61 ± 1.77 mL/g VS) (Fig. 1d). Among these, CO_2-NBW showed the highest methane production, with its lower pH creating acidic conditions during straw anaerobic digestion, promoting the hydrolysis and acidogenesis processes, and providing substrates for the hydrogenotrophic methanogenesis pathway, ultimately achieving the highest methane production. This finding is consistent with previous studies that demonstrated the effectiveness of

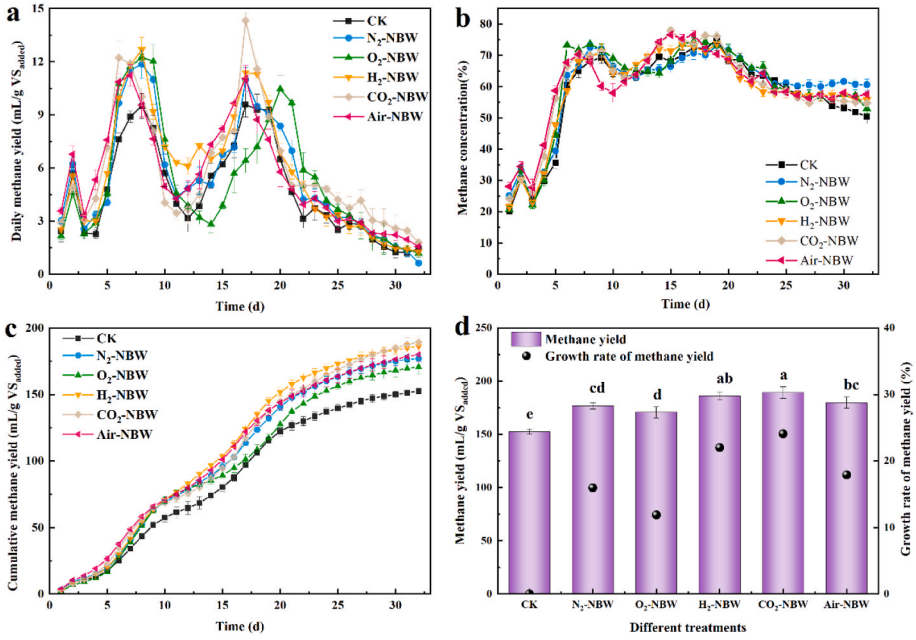


Fig. 1. (a) Daily methane yield; (b) Methane concentration; (c) Cumulative methane yield; (d) Differences in cumulative methane yield. Different letters indicated statistical differences ($p < 0.05$).

CO_2-enriched nanobubble water in enhancing methanogenesis by stimulating hydrolysis and shifting the methanogenic pathway toward hydrogenotrophic methanogenesis [19,25]. Similarly, H_2-NBW, by providing H_2 as a substrate for hydrogenotrophic methanogenesis, also resulted in higher methane production. This finding is consistent with those of other researchers [22]. A significant methane yield improvement was also observed in the AD system using Air-NBW. Although the presence of air usually leads to the death of methane-producing bacteria, under actual AD conditions, the limited oxygen is quickly consumed by aerobic microorganisms or facultative anaerobes, rapidly transitioning the reaction environment to anaerobic [29]. In addition, some methane-producing bacteria have a certain tolerance to oxygen, and after the inhibition from oxygen exposure is removed, their biological activity can quickly recover. Therefore, Air-NBW has a significantly greater promoting effect on AD compared to CK with distilled water. The presence of limited oxygen can stimulate the proliferation of lignocellulose-degrading microorganisms, thereby accelerating the breakdown of lignocellulose—a key factor in enhancing the hydrolysis efficiency of anaerobic digestion. Furthermore, N_2-NBW helps maintain a strict anaerobic environment, preventing the inactivation of anaerobic bacteria due to fluctuations in dissolved oxygen and improving methane production by enhancing mass transfer efficiency. With long-term use, N_2-NBW can provide more stable conditions for AD, further contributing to improved methane production.

To more comprehensively analyze the kinetic characteristics of the anaerobic fermentation process with different types of NBW or distilled water (DW), this study applied the modified Gompertz model, Logistic model, and first-order hydrolysis kinetic model to fit the data, with relevant parameters shown in Table 2. The modified Gompertz and Logistic models are widely applied to describe cumulative methane production because they can capture the sigmoidal pattern of biogas generation and provide reliable estimates of the methane production potential, maximum production rate, and lag phase. In contrast, the first-order hydrolysis kinetic model assumes that substrate hydrolysis is the rate-limiting step of anaerobic digestion and thus focuses on the degradation kinetics of organic matter. The combined use of these models allows both empirical fitting of methane production behavior and mechanistic interpretation of substrate degradation. The results

Table 2
Fitting of methane production kinetic curves using the modified gompertz model, logistic model, and kinetic modeling of primary hydrolysis.

Kinetics model	Model parameters	Treatment					
		CK	N_2-NBW	O_2-NBW	H_2-NBW	CO_2-NBW	Air-NBW
Measured value	B_{max}(mL/g VS)	152.61 ± 1.77^{e}	176.94 ± 3.27^{cd}	170.72 ± 5.14^{d}	186.17 ± 3.31^{ab}	189.36 ± 5.32^{a}	179.94 ± 5.22^{bc}
Modified Gompertz model	P_{max} (mL/g VS)	167.60 ± 3.71^{c}	174.76 ± 4.79^{c}	170.28 ± 5.04^{c}	187.75 ± 5.90^{b}	202.92 ± 6.11^{a}	176.00 ± 6.15^{c}
	R_{max} (mL/g VS/d)	7.52 ± 0.14^{d}	9.96 ± 0.28^{ab}	9.36 ± 0.32^{c}	10.26 ± 0.24^{a}	9.46 ± 0.19^{bc}	9.76 ± 0.26^{bc}
	λ(d)	3.29 ± 0.09^{c}	3.14 ± 0.13^{d}	3.12 ± 0.20^{d}	3.37 ± 0.14^{b}	3.85 ± 0.20^{a}	2.92 ± 0.17^{e}
	R^2	0.9966	0.9923	0.9951	0.9927	0.9966	0.9901
	Diff(%)	−1.94[a]	1.25	0.26	−0.84[a]	−6.68[a]	2.24
Logistic model	P_{max}(mL/g VS)	151.78 ± 2.99^{c}	150.14 ± 5.76^{c}	149.93 ± 4.15^{c}	163.53 ± 7.78^{ab}	174.61 ± 4.81^{a}	154.84 ± 7.57^{bc}
	k	0.24 ± 0.01^{c}	0.35 ± 0.02^{a}	0.31 ± 0.02^{b}	0.31 ± 0.02^{b}	0.27 ± 0.01^{c}	0.32 ± 0.02^{ab}
	to	14.01 ± 0.36^{a}	10.19 ± 0.37^{c}	11.66 ± 0.55^{b}	11.29 ± 0.43^{b}	13.71 ± 0.46^{a}	11.01 ± 0.50^{bc}
	R^2	0.9937	0.9715	0.9850	0.9713	0.9906	0.9647
	Diff(%)	−0.54[a]	−15.15[a]	−12.18[a]	−12.16[a]	−7.79[a]	−13.95[a]
Kinetic modeling of primary hydrolysis	K(d^{-1})	0.0262 ± 0.0020^{d}	0.0368 ± 0.0023^{c}	0.0450 ± 0.0027^{b}	0.0383 ± 0.0026^{c}	0.0509 ± 0.0034^{a}	0.0416 ± 0.0024^{bc}
	R^2	0.8408	0.9118	0.9453	0.8960	0.9461	0.9383

[a] The negative value means that the predicted value is lower than the measured value. Diff means difference.

showed that the modified Gompertz model had a better fit (R^2 = 0.9901–0.9966) compared to the Logistic model (R^2 = 0.9647–0.9937) and the first-order hydrolysis kinetic model (R^2 = 0.8408–0.9461). This model was used to fit the cumulative methane production, and the obtained parameters included the final methane production (Pmax), maximum methane production rate (Rmax), and lag phase (λ). The results indicated that the predicted final methane production was in close agreement with the actual measured values. The predicted final methane production closely matched the actual measured values, confirming the model's reliability.The lag phase parameter (λ) represents the initial period before microorganisms adapt to the new environment, proliferate, and enter the exponential growth phase. Generally, a larger λ indicates slower microbial adaptation, delayed enzymatic activity, or the time required to remove inhibitory compounds, resulting in a slower onset of methane production [30]. Comparing λ values among treatments, DW exhibited a lag phase of 3.29 days, while O_2-NBW (3.12 days) and Air-NBW (2.92 days) showed shorter lag phases, suggesting that these treatments accelerated microbial adaptation and enzyme activation. The presence of oxygen in O_2-NBW and Air-NBW likely promotes electron transfer, stimulates reactive oxygen species generation, and enhances the activity of hydrolytic and acidogenic enzymes, facilitating faster substrate breakdown [29]. In contrast, H_2-NBW (3.37 days) and CO_2-NBW (3.85 days) displayed longer lag phases, indicating slower initial fermentation. This may result from transient inhibitory effects of dissolved hydrogen or carbon dioxide on specific microbial populations, which delay the establishment of active methanogenic consortia, although these gases could enhance methane production in later stages. Overall, these findings suggest that NBW not only accelerates methane production but also modulates microbial community dynamics and functional metabolism during the lag phase. The type of gas and its interactions with microbial adaptation and environmental conditions play key roles in determining the duration of the lag phase and the subsequent efficiency of anaerobic digestion.

3.2. Changes in VFAs, pH, SCOD, and TS/VS removal rates during AD

In the AD system, pH is a comprehensive reflection of various biochemical reactions, including CO_2 dissolution equilibrium, acid-base balance, and ion dissolution equilibrium [31]. According to the literature, the pH range suitable for the growth and metabolism of methane-producing bacteria is 6.5–8.2. The pH values of all experimental groups in the first two days ranged from 6.68 to 7.02, which were lower than the subsequent pH levels. This indicates that during this period, due to the microbial hydrolysis and acidogenesis processes, soluble substances and lignocellulose were decomposed, producing lactic acid, which was gradually converted into short-chain fatty acids, primarily including formic acid, acetic acid, and propionic acid (Fig. 2a). During a typical methane production process, short-chain volatile fatty acids are progressively consumed by methanogenic archaea, resulting in a gradual decline in their concentrations. In the experiment, the TVFA concentrations in the CK and treatment groups ranged from 295.33 to 5741.34 mg/L and 378.12–8592.67 mg/L, respectively. The incorporation of NBW facilitates the degradation of organic compounds in the straw and promotes the formation of soluble substances, thereby significantly enhancing acid production efficiency [32]. Notably, the TVFA concentration in the CO_2-NBW treatment group (452.10–8784.13 mg/L) was significantly higher than that in the other experimental groups, and its pH was lower than that of the others due to high concentrations of VFAs. This enhanced the metabolic activity in the acidogenesis phase, accelerated the acidification reaction, and consequently lowered the pH of the system [23]. This mechanism suggests that CO_2-NBW can effectively enhance hydrolysis and acid production efficiency by regulating the acidification environment, providing a new approach for optimizing the AD process [25].

SCOD is typically used to evaluate substrate solubility during the hydrolysis process [33]. The trend in SCOD changes was similar to that of TVFA concentrations (Fig. 2b), with all treatment groups showing an initial increase followed by a decrease. On day 5, SCOD reached its maximum value, which may have been due to the release of a large amount of volatile fatty acids (VFA) during the hydrolysis phase. However, at this point, the SCOD in the NBW treatment groups was lower than that in the DW control group, indicating that after adding NBW, soluble organic matter was consumed more rapidly by microorganisms in the AD process, being used for metabolism and gas production. Starting from day 5, the SCOD in all treatment groups gradually decreased, further indicating that the soluble organic matter had been effectively utilized by the digestion system, converting it into methane and other metabolic products [34].

TS and VS removal rates are important indicators for assessing the degree of substrate degradation. Typically, the degradation rates of TS

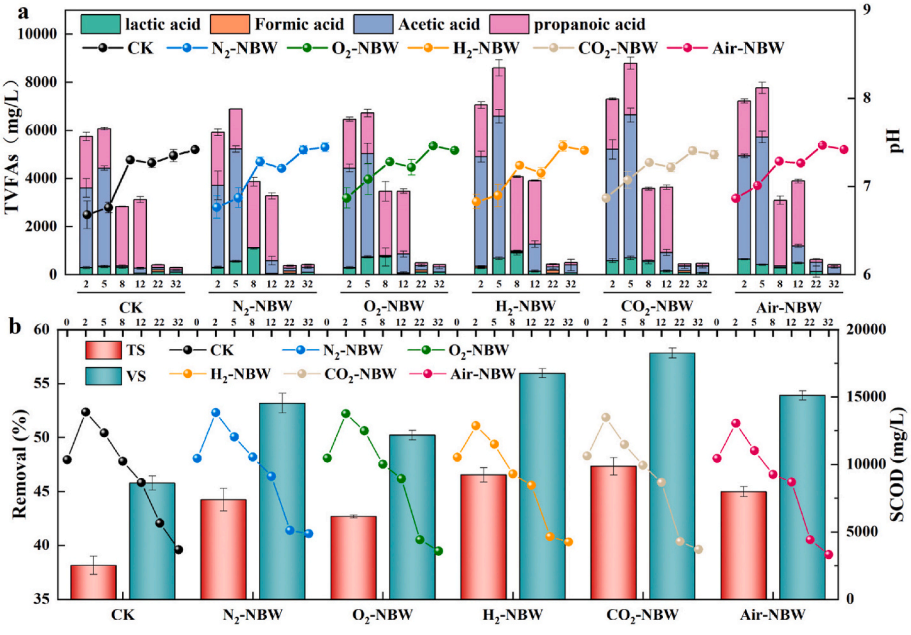


Fig. 2. The variation in (a) TVFAs and pH; (b) TS and VS Degradation Rates and SCOD.

and VS are positively correlated with methane production [35]. As shown in Fig. 2b, the TS and VS removal rates in the treatment groups ranged from 50.22 % to 57.81 % and 42.68 %–47.34 %, respectively, which were significantly higher than those in the CK (TS: 45.78 %, VS: 38.15 %) ($p < 0.05$). Among these, the CO_2-NBW treatment group exhibited the highest TS and VS removal rates, at 57.81 % and 47.34 %, respectively, which were 26.28 % and 24.09 % higher than those in the CK. This result is consistent with the highest cumulative methane production observed in this group (Fig. 2b) and is also in agreement with previous studies reporting that CO_2 nanobubble water can enhance organic matter degradation and methane yield during anaerobic digestion [23]. In summary, adding NBW can increase the substrate's hydrolysis and acidogenesis efficiency, which will raise the rate of VS degradation and, eventually, methane production [36].

3.3. Effects of NBW on microbial community structure and metabolic functions

High-throughput sequencing analysis was conducted to further explore the mechanisms of NBW and to elucidate the microbial community structure changes under different NBW treatments. The results of the Alpha diversity analysis for bacteria and archaea are shown in Table 3, with coverage indices exceeding 0.99, indicating that the sequencing data adequately represent the true diversity of the microbial communities in the samples. The diversity of the biological communities was evaluated using the Shannon and Simpson indices [37], whilst the microbial community richness was reflected using the Ace and Chao indices [38].

In the bacterial sequencing, the Sobs index, Chao index, and Ace index indicated that the microbial community abundance in O_2-NBW and Air-NBW was higher than the CK and N_2-NBW. This may be because oxygen promotes the growth of certain facultative anaerobic microorganisms, suggesting that trace amounts of oxygen are beneficial for improving the performance of bacterial microbial communities in the AD process. Compared to the bacterial community, the richness index of archaea was significantly lower [39]. The Sobs value in the CK was the lowest (20), while the Sobs values for CO_2-NBW and H_2-NBW were the highest (28 and 26, respectively). In environments with carbon dioxide and hydrogen (CO_2-NBW, H_2-NBW), the richness of archaea was slightly higher.

The results of the Shannon and Simpson indices indicate that, in the bacterial community, aerobic environments (such as Air-NBW and O_2-NBW) promoted the diversity of the community while increasing evenness. This suggests that different types of bacteria are able to coexist and grow together in these environments [40]. In the archaeal community, the carbon dioxide environment was beneficial for increasing both diversity and evenness, while the hydrogen environment led to certain archaeal species dominating. This may be due to the strong adaptability and competitiveness of specific archaea to hydrogen.

3.3.1. Dominant bacterial community

The dominant phyla are *Bacteroidota* and *Firmicutes*, which are also the two most common phyla in AD (Fig. 3a) [41]. *Bacteroidota* dominates the hydrolysis and primary fermentation stages, responsible for converting complex organic matter (such as polysaccharides and proteins) into simple small molecules and volatile fatty acids [42]. The highest abundance of *Bacteroidota* was observed in CO_2-NBW, suggesting that CO_2 may influence the pH environment of the system, thereby regulating the microbial community structure. *Firmicutes* include many strict anaerobes and hydrogen-producing bacteria, which can utilize hydrogen as an electron donor to participate in metabolism [43]. These bacteria produce cellulases, proteases, and various extracellular enzymes to degrade complex organic matter [44]. H_2-NBW, by releasing a high concentration of dissolved hydrogen, provides abundant hydrogen resources, significantly promoting the growth and metabolic activity of these bacterial species. As a result, this phylum had the highest abundance in H_2-NBW. At the genus level (Fig. 3b), *Ruminofilibacter* had the highest abundance, and it is typically found in the rumen of ruminant animals. *Ruminofilibacter* is capable of degrading cellulose and converting it into short-chain fatty acids [45]. In CO_2-NBW, *Ruminofilibacter* had the highest proportion, possibly directly participating in the cellulose degradation metabolic pathway, promoting substrate conversion and microbial growth. It may also collaborate with hydrogen-producing bacteria to provide sufficient substrates for methane-producing bacteria ($CO_2 + H_2 \rightarrow CH_4$), forming a mutualistic relationship. This metabolic network may further enhance the activity of *Ruminofilibacter*.

3.3.2. Dominant Archaeal Community

Compared to the bacterial community, the archaeal community exhibited significantly lower diversity and richness. During the methanogenesis stage, different archaea play key roles in methane production. At the phylum level, *Halobacterota* and *Euryarchaeota* dominated, with their combined relative abundance accounting for 96.95 %–99.95 % of the total relative abundance at the phylum level (Fig. 3c). At the genus level, *Methanosarcina* and *Methanobacterium* were the dominant genera (Fig. 3d). *Methanosarcina* is a commonly observed acetoclastic methanogen in methane production systems and was enriched across all groups. In addition, *Methanobacterium* is a hydrogenotrophic methanogen whose primary energy sources are CO_2 and H_2 [46]. Its enrichment was higher in the CO_2-NBW group compared to the CK, indicating that CO_2-NBW primarily relies on the hydrogenotrophic methanogenesis pathway.

To further evaluate the metabolic responses mediated by different NBWs during AD of straw, the relative abundances of key enzyme genes using KEGG metabolic pathways (Fig. 4). In the hydrolysis-acidogenesis stage (Fig. 4a), the enzyme levels across the groups were observed (Fig. 4b). Compared to the CK, CO_2-NBW showed a decrease in cellulase (EC: 3.2.1.4) involved in the hydrolysis of cellulose to glucose, while β-glucosidase (EC:3.2.1.21) increased. Similarly, xylanase (EC: 3.2.1.8) was involved in the hydrolysis of hemicellulose to xylulose decreased,

Table 3
Alpha diversity analysis of bacteria and archaea.

Sample		Sobs[a]	ACE[a]	Chao[a]	Shannon[b]	Simpson[b]	Coverage (100 %)[c]
Bacteria	CK	595.00	621.97	614.01	4.45	0.03	99.82
	N_2-NBW	548.00	549.86	548.19	4.38	0.03	99.98
	O_2-NBW	658.00	686.67	677.68	4.80	0.02	99.81
	H_2-NBW	570.00	570.00	570.00	4.56	0.02	100.00
	CO_2-NBW	668.00	678.89	670.95	4.57	0.03	99.91
	Air-NBW	746.00	773.12	759.21	4.61	0.03	99.81
Archaea	CK	20.00	20.00	20.00	1.44	0.34	100.00
	N_2-NBW	23.00	23.00	23.00	1.38	0.35	100.00
	O_2-NBW	26.00	26.00	26.00	1.40	0.34	100.00
	H_2-NBW	26.00	27.09	26.33	0.87	0.64	99.99
	CO_2-NBW	28.00	28.00	28.00	1.45	0.32	100.00
	Air-NBW	24.00	24.00	24.00	1.10	0.52	100.00

Note: a) Community richness index; b) Community diversity index; c) Coverage index.

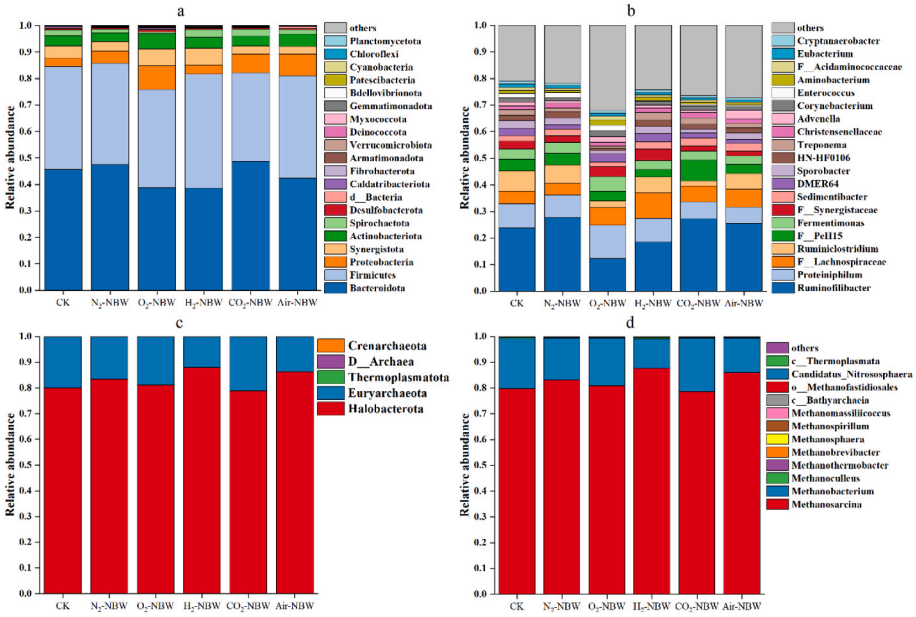


Fig. 3. Analysis of Bacterial and Archaeal Communities: (a) Bacterial community at the phylum level; (b) Bacterial community at the genus level; (c) Archaeal community at the phylum level; (d) Archaeal community at the genus level.

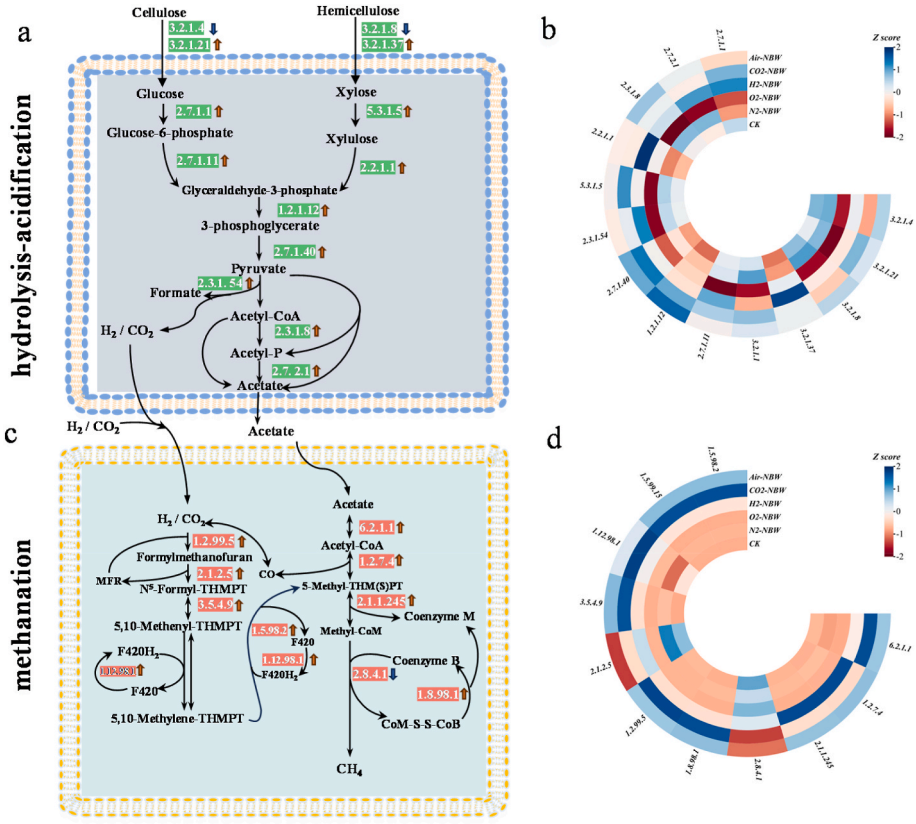


Fig. 4. Metabolic Pathways of Hydrolysis and Acidogenesis (a) and Methanogenesis (c) During Straw AD, and Heatmaps of the Relative Abundance of Functional Enzymes Involved in the Corresponding Pathways Across Experimental Groups (b, d).

while xylosidase (EC:3.2.1.37) increased. This is likely because, in the later stages of fermentation, complex polysaccharides (cellulose and hemicellulose) are gradually depleted, leading to reduced activity of primary degrading enzymes (cellulase and xylanase). Conversely, the activity of enzymes degrading intermediate products (e.g., cellobiose and xylobiose), such as β-glucosidase and xylosidase, increased. CO_2-NBW may have optimized the fermentation environment (e.g., oxygen conditions, pH balance, or enzyme activity), enabling microbial metabolism to more efficiently shift toward utilizing intermediate degradation products like glucose and xylose. Additionally, during the acidogenesis phase, CO_2-NBW exhibited a higher abundance of enzymes compared to the CK, including hexokinase (EC:2.7.1.1), adenylate

kinase (EC:2.7.1.11), xylose isomerase (EC:5.3.1.5), transketolase (EC:2.2.1.1), glyceraldehyde-3-phosphate dehydrogenase (EC:1.2.1.12), pyruvate kinase (EC:2.7.1.40), formate C-acetyltransferase (EC:2.3.1.54), phosphate acetyltransferase (EC:2.3.1.8), and acetate kinase (EC:2.7.2.1). This suggests that limiting the rates of hydrolysis and acid formation can aid in avoiding the buildup of acid during anaerobic fermentation, guaranteeing the effective and steady functioning of the anaerobic digestion process [46].

During the methanogenesis stage (Fig. 4c), most enzymes were more abundant in the CO_2-NBW group compared to the CK (Fig. 4d). These enzymes include formylmethanofuran dehydrogenase (EC:1.2.99.5), glutamate formimidoyltransferase (EC:2.1.2.5), methenyltetrahydrofolate cyclohydrolase (EC:3.5.4.9), coenzyme F420 hydrogenase (EC:1.12.98.1), acetyl-CoA synthase (EC:6.2.1.1), anaerobic carbon monoxide dehydrogenase (EC:1.2.7.4), S-adenosyl-L-methionine: 3-methyl-2-oxobutanoate methyltransferase (EC: 2.1.1.245), and heterodisulfide reductase (EC:1.8.98.1). Notably, the abundance of adenosine triphosphate sulfurylase (EC:2.8.4.1) was lower in the CO_2-NBW and H_2-NBW treatment groups compared to the CK, suggesting that this corresponds to the later stage of methanogenesis. The high activity of heterodisulfide reductase (EC:1.8.98.1) facilitated the rapid reduction of heterodisulfide coenzyme (CoM-S-S-CoB) to its active forms (HS-CoM and HS-CoB) by consuming reducing equivalents such as H_2 and $FADH_2$. This process ensured the smooth operation of the methanogenesis cycle, allowing adenosine triphosphate sulfurylase (EC:2.8.4.1), despite its lower abundance, to efficiently utilize substrates for methane production [47].

3.4. Preliminary analysis of the promoting mechanisms of different NBWs

Different types of NBW, through their unique gas dissolution characteristics and gas-liquid interface behavior, promote the hydrolysis, acidogenesis, and methanogenesis stages. Among them, CO_2-NBW exerts the most significant promoting effect by regulating the acidification environment, providing carbon sources, and promoting the enrichment of hydrogenotrophic methanogens like *Methanobacterium*. H_2-NBW, by providing substrates for the hydrogenotrophic pathway, also has a strong advantage in the methanogenesis phase. In contrast, N_2-NBW, O_2-NBW, and Air-NBW primarily affect the hydrolysis and acidogenesis stages in the early fermentation phase. These observations suggest that the role of NBW is not limited to simply enhancing gas–liquid mass transfer, but also involves modifying the microbial community structure and the metabolic balance among different functional groups. For instance, the stimulation of hydrogenotrophic methanogens under CO_2-NBW and H_2-NBW treatments indicates that nanobubbles may act as selective regulators, favoring specific methanogenic pathways. Similarly, the improved hydrolysis and acidogenesis under O_2-NBW and Air-NBW imply that small amounts of oxygen could enhance extracellular enzyme activity and accelerate the breakdown of complex substrates. Although preliminary, these findings provide mechanistic clues as to why different gases lead to divergent promoting effects. They also highlight that the choice of NBW type could be tailored to target specific stages of anaerobic digestion, thereby offering a flexible strategy for process optimization. These mechanisms provide diversified regulatory strategies for optimizing straw anaerobic fermentation and offer theoretical support for NBW selection and process improvements in practical applications. Future studies should examine how substrate heterogeneity, operational parameters (e.g., organic loading rate, temperature, pH), and potential non-biological effects (e.g., mass transfer, micromixing, mechanical disruption) influence AD, as well as their impacts on effluent quality, nutrient balance, and downstream process stability, to fully assess the practical applicability of NBW.

4. Conclusion

In this study, NBWs prepared with different gas types all enhanced the hydrolysis-acidogenesis degree and degradation rate of straw, and optimized the functional microbial community structure in hydrolysis, acidogenesis, and methanogenesis, thereby improving the methane production efficiency from straw. Among them, the cumulative methane production in the N_2-NBW, O_2-NBW, H_2-NBW, CO_2-NBW, and Air-NBW treatment groups increased by 11.86 %–24.08 % compared to the CK. The TS and VS removal rates increased by 9.70 %–26.28 % and 11.87 %–24.09 %, respectively. CO_2-NBW showed the best results with maximum methane reaching 189.36 ± 5.32 mL/g VS, specifically enhancing the expression of enzymes involved in hydrogenotrophic methanogenesis. In conclusion, utilizing nanobubble technology to stabilize different gases for generating NBWs can enhance methane production in corn straw AD. The purpose of this study is to offer some references for the use of nanobubble technology in AD to promote the synthesis of methane from straw biomass.

CRediT authorship contribution statement

Chuanfei Bian: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Hao Liu:** Data curation. **Xin Gao:** Data curation. **Shuaishuai Ma:** Data curation. **Xiurong Ren:** Data curation. **Muhammad Usman Khan:** Writing – original draft. **Luyao Wei:** Data curation. **Xufeng Yuan:** Data curation. **Hongliang Wang:** Supervision, Resources, Funding acquisition, Data curation. **Wanbin Zhu:** Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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